

smallest (more negative), hence  $E_k$  and speed is the largest.

2 (a) (i) By  $\frac{pV}{T} = \text{constant}$ ,

$$\begin{aligned}\frac{p_C V_C}{T_C} &= \frac{p_B V_B}{T_B} \\ p_C &= \frac{p_B V_B}{T_B} \times \frac{T_C}{V_C} \\ &= \frac{(7.0 \times 10^5) \times 0.0015}{1260} \times \frac{960}{0.0040} \\ &= 2.00 \times 10^5 \text{ Pa}\end{aligned}$$

(ii)  $W = \text{Area under the graph}$

$$\begin{aligned}&= \frac{1}{2} (7.0 + 2.0) \times 10^5 \times (4.0 - 1.5) \times 10^{-3} \\ &= 1130 \text{ J}\end{aligned}$$

- (b) 1. The process is carried out slowly.  
2. The walls of the engine are good conductors of heat.  
3. The engine is surrounded by a heat reservoir at constant temperature.  
Maximum 2 marks

(c)

| change | work done on gas / J | heat supplied to gas / J | increase in internal energy / J |
|--------|----------------------|--------------------------|---------------------------------|
| A → B  | -1000                | 2625                     | 1625                            |
| B → C  | -1125                | 500                      | -625                            |
| C → D  | 0                    | -1000                    | -1000                           |
| D → A  | 555                  | -555                     | 0                               |

- (d) During an explosion, the pressure at different points of the gas is not uniform. Hence, pressure is undefined and cannot be represented as a point or a curve on the  $p$ - $V$  diagram.

3 (a) For  $a$  to be minimum, the minimum intensity detected directly below it must be the